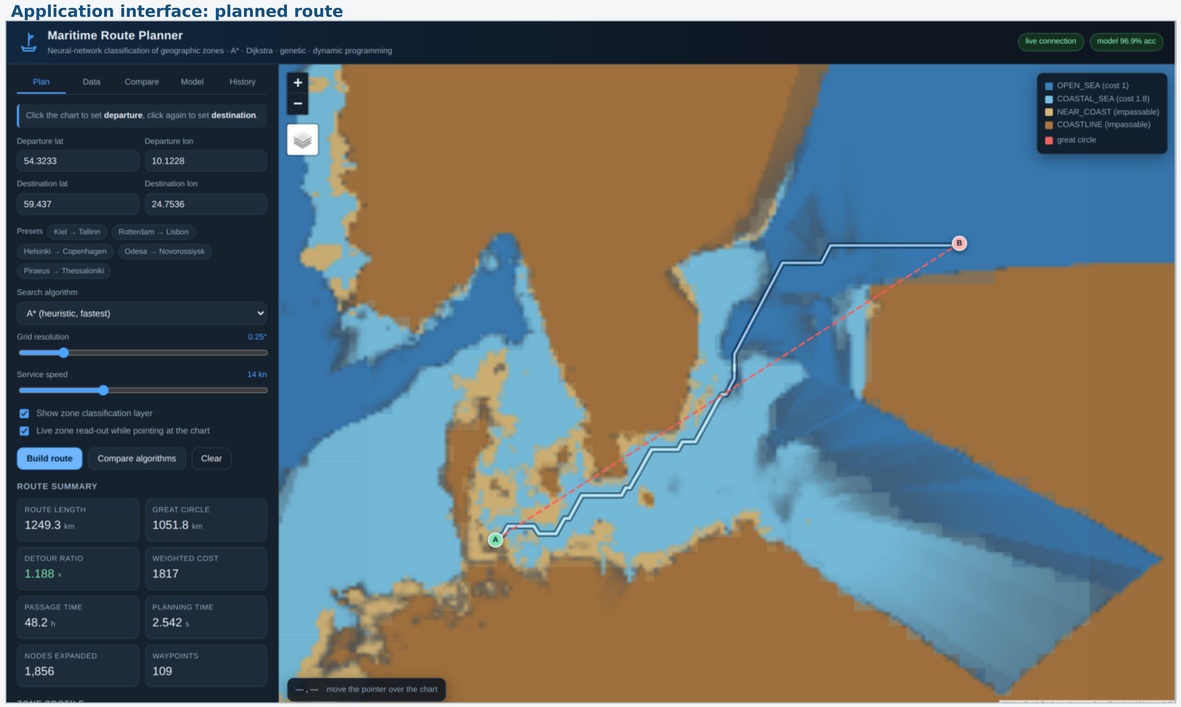
Test results

Interface, classification quality, planned routes and comparison with the shortest-distance route

Sheet 3

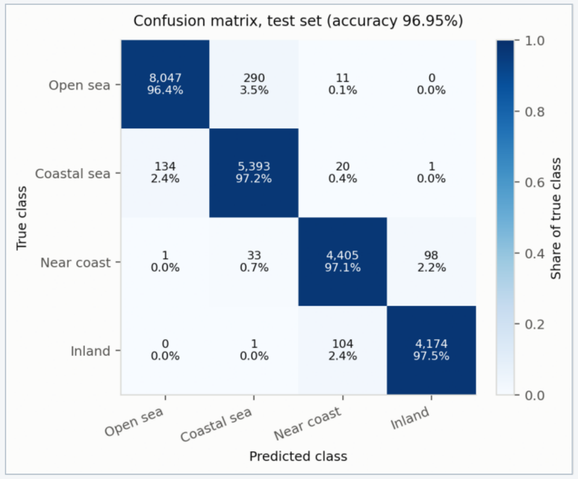
Format A1

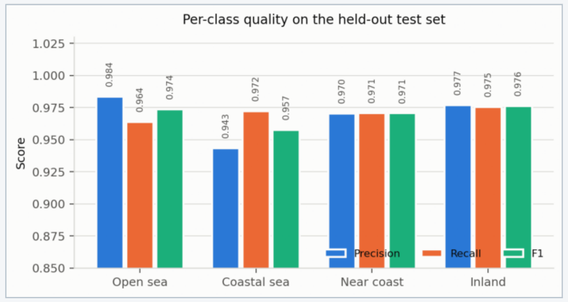
Application interface: planned route



Kiel → Tallinn, A*, lattice 0.25°: the route stays in navigable zones while the great circle (red) crosses Denmark and Sweden

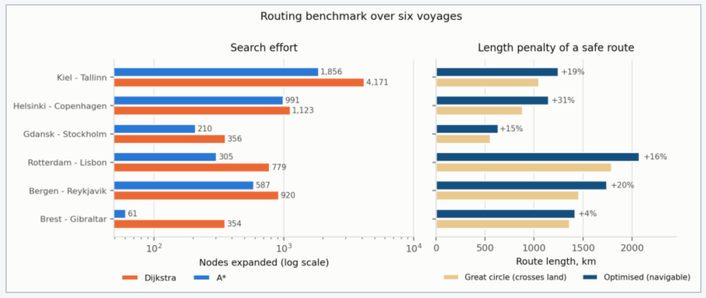
Classification quality on the test set





Metric	Value
Test points	22,712
Accuracy	96.95 %
Macro F1	0.9695
Weighted F1	0.9695
Cohen κ	0.9582
Macro ROC-AUC	0.9985
Mean confidence	0.9504

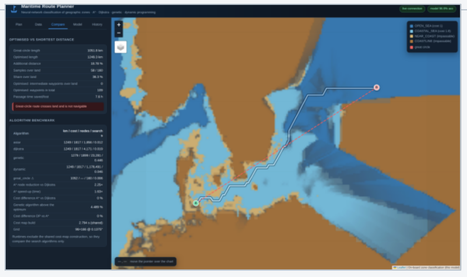
Routing benchmark



Voyage	A* km	Great circle km	Extra	A* nodes	Dijkstra nodes	Search s
Kiel - Tallinn	1,249	1,052	+18.8 %	1,856	4,171	0.012
Helsinki - Copenhagen	1,152	883	+30.5 %	991	1,123	0.007
Gdansk - Stockholm	636	555	+14.6 %	210	356	0.003
Rotterdam - Lisbon	2,077	1,794	+15.8 %	305	779	0.004
Bergen - Reykjavik	1,746	1,458	+19.8 %	587	920	0.006
Brest - Gibraltar	1,423	1,364	+4.3 %	61	354	0.003

Across 6 voyages A* expands 2.5× fewer nodes than Dijkstra and runs 1.233× faster, while both return the identical optimal cost (max deviation 0.0 %). All 6 great-circle routes cross land; the optimised routes are 17.3 % longer on average and fully navigable.

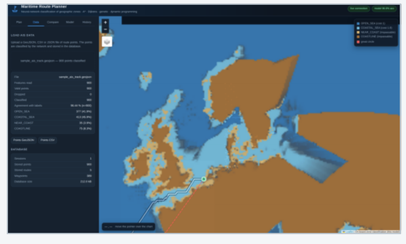
Comparison, uploaded data and stored routes



Algorithm comparison on one shared cost map



Rotterdam → Lisbon, lattice 0.25°



AIS upload: 900 points classified against their labels



Model panel: metrics, matrix, training curves

Application for constructing a maritime vessel's course based on neural-network classification of geographic zones

Diploma project · Student: Li Haoran

Brest State Technical University · Faculty EIS, Department IIT · 2026/27